COHEN-MACAULAYNESS AT A POINT IS EQUIVALENT TO THE EXISTENCE OF A FLAT QUASI-FINITE MAP TO AFFINE SPACE

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Editorial note. Statement span is lines 6013-6030 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 6032-6065, the complete author proof (32 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Cohen-Macaulay morphisms, quasi-finite morphisms to affine space, and the fibres' dimension; the construction is retained.

Source reference. Original lemma 00QE: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6065–6065. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00RE: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6065–6065. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma theorem-criterion-flatness-fibre: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 4590–4599. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma theorem-openness-flatness: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 4446–4449. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes which is flat and locally of finite presentation. Let $x \in X$ with image $s \in S$. Set $d = \dim_x(X_s)$. The following are equivalent*

(1) *f is Cohen-Macaulay at x,*
(2) *there exists an open neighbourhood $U \subset X$ of x and a locally quasi-finite morphism $U \to \mathbf{A}^d_S$ over S which is flat at x,*

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(3) *there exists an open neighbourhood $U \subset X$ of x and a locally quasi-finite flat morphism $U \to \mathbf{A}^d_S$ over S,*
(4) *for any S-morphism $g : U \to \mathbf{A}^d_S$ of an open neighbourhood $U \subset X$ of x we have: g is quasi-finite at $x \Rightarrow g$ is flat at x.*

Proof. Openness of flatness shows (2) and (3) are equivalent, see Theorem theorem-openness-flatness.

Choose affine open $U = \mathrm{Spec}(A) \subset X$ with $x \in U$ and $V = \mathrm{Spec}(R) \subset S$ with $f(U) \subset V$. Then $R \to A$ is a flat ring map of finite presentation. Let $\mathfrak{p} \subset A$ be the prime ideal corresponding to x. After replacing A by a principal localization we may assume there exists a quasi-finite map $R[x_1, \ldots, x_d] \to A$, see Algebra, Lemma 00QE. Thus there exists at least one pair (U, g) consisting of an open neighbourhood $U \subset X$ of x and a locally[1] quasi-finite morphism $g : U \to \mathbf{A}^d_S$.

Claim: Given $R \to A$ flat and of finite presentation, a prime $\mathfrak{p} \subset A$ and $\varphi : R[x_1, \ldots, x_d] \to A$ quasi-finite at $\mathfrak{p}$ we have: $\mathrm{Spec}(\varphi)$ is flat at $\mathfrak{p}$ if and only if $\mathrm{Spec}(A) \to \mathrm{Spec}(R)$ is Cohen-Macaulay at $\mathfrak{p}$. Namely, by Theorem theorem-criterion-flatness-fibre flatness may be checked on fibres. The same is true for being Cohen-Macaulay (as A is already assumed flat over R). Thus the claim follows from Algebra, Lemma 00RE.

The claim shows that (1) is equivalent to (4) and combined with the fact that we have constructed a suitable (U, g) in the second paragraph, the claim also shows that (1) is equivalent to (2). □

[1] If S is quasi-separated, then g will be quasi-finite.